

ANSI SQL Using MySQL – Exercises 1 to 25 Answers

1. User Upcoming Events

```
SELECT e.title, e.city, e.start_date FROM Events e JOIN Registrations r ON e.event_id = r.event_id JOIN Users u ON r.user_id = u.user_id WHERE e.status='upcoming' AND e.city=u.city ORDER BY e.start_date;
```

2. Top Rated Events

```
SELECT e.event_id,e.title,AVG(f.rating) AS avg_rating FROM Events e JOIN Feedback f ON e.event_id=f.event_id GROUP BY e.event_id,e.title HAVING COUNT(f.feedback_id)>=10 ORDER BY avg_rating DESC;
```

3. Inactive Users

```
SELECT * FROM Users WHERE user_id NOT IN (SELECT DISTINCT user_id FROM Registrations WHERE registration_date >= CURDATE()-INTERVAL 90 DAY);
```

4. Peak Session Hours

```
SELECT e.title, COUNT(s.session_id) AS peak_sessions FROM Events e LEFT JOIN Sessions s ON e.event_id=s.event_id WHERE TIME(s.start_time) BETWEEN '10:00:00' AND '12:00:00' GROUP BY e.title;
```

5. Most Active Cities

```
SELECT u.city, COUNT(DISTINCT r.user_id) AS registrations FROM Users u JOIN Registrations r ON u.user_id=r.user_id GROUP BY u.city ORDER BY registrations DESC LIMIT 5;
```

6. Event Resource Summary

```
SELECT e.title, COUNT(CASE WHEN resource_type='pdf' THEN 1 END) AS pdf_count, COUNT(CASE WHEN resource_type='image' THEN 1 END) AS image_count, COUNT(CASE WHEN resource_type='link' THEN 1 END) AS link_count FROM Events e LEFT JOIN Resources r ON e.event_id=r.event_id GROUP BY e.title;
```

7. Low Feedback Alerts

```
SELECT u.full_name,e.title,f.rating,f.comments FROM Feedback f JOIN Users u ON f.user_id=u.user_id JOIN Events e ON f.event_id=e.event_id WHERE f.rating < 3;
```

8. Sessions per Upcoming Event

```
SELECT e.title, COUNT(s.session_id) AS session_count FROM Events e LEFT JOIN Sessions s ON e.event_id=s.event_id WHERE e.status='upcoming' GROUP BY e.title;
```

9. Organizer Event Summary

```
SELECT u.full_name,e.status,COUNT(*) AS total_events FROM Events e JOIN Users u ON  
e.organizer_id=u.user_id GROUP BY u.full_name,e.status;
```

10. Feedback Gap

```
SELECT e.title FROM Events e JOIN Registrations r ON e.event_id=r.event_id LEFT JOIN Feedback f ON  
e.event_id=f.event_id WHERE f.feedback_id IS NULL GROUP BY e.title;
```

11. Daily New User Count

```
SELECT registration_date, COUNT(*) AS user_count FROM Users WHERE registration_date >=  
CURDATE()-INTERVAL 7 DAY GROUP BY registration_date;
```

12. Event with Maximum Sessions

```
SELECT e.title, COUNT(s.session_id) AS session_count FROM Events e JOIN Sessions s ON  
e.event_id=s.event_id GROUP BY e.event_id,e.title HAVING COUNT(s.session_id)=(SELECT MAX(cnt)  
FROM (SELECT COUNT(*) cnt FROM Sessions GROUP BY event_id) t);
```

13. Average Rating per City

```
SELECT e.city, AVG(f.rating) AS avg_rating FROM Events e JOIN Feedback f ON e.event_id=f.event_id  
GROUP BY e.city;
```

14. Most Registered Events

```
SELECT e.title, COUNT(r.registration_id) AS total_registrations FROM Events e JOIN Registrations r ON  
e.event_id=r.event_id GROUP BY e.title ORDER BY total_registrations DESC LIMIT 3;
```

15. Event Session Time Conflict

```
SELECT s1.event_id,s1.title,s2.title FROM Sessions s1 JOIN Sessions s2 ON s1.event_id=s2.event_id AND  
s1.session_id<s2.start_time;
```

16. Unregistered Active Users

```
SELECT * FROM Users u WHERE u.registration_date >= CURDATE()-INTERVAL 30 DAY AND NOT  
EXISTS (SELECT 1 FROM Registrations r WHERE r.user_id=u.user_id);
```

17. Multi-Session Speakers

```
SELECT speaker_name, COUNT(*) AS session_count FROM Sessions GROUP BY speaker_name  
HAVING COUNT(*) > 1;
```

18. Resource Availability Check

```
SELECT e.title FROM Events e LEFT JOIN Resources r ON e.event_id=r.event_id WHERE r.resource_id IS NULL;
```

19. Completed Events with Feedback Summary

```
SELECT e.title, COUNT(DISTINCT r.registration_id) AS total_registrations, AVG(f.rating) AS avg_rating  
FROM Events e LEFT JOIN Registrations r ON e.event_id=r.event_id LEFT JOIN Feedback f ON  
e.event_id=f.event_id WHERE e.status='completed' GROUP BY e.title;
```

20. User Engagement Index

```
SELECT u.full_name, COUNT(DISTINCT r.event_id) AS attended_events, COUNT(DISTINCT  
f.feedback_id) AS feedback_count FROM Users u LEFT JOIN Registrations r ON u.user_id=r.user_id LEFT  
JOIN Feedback f ON u.user_id=f.user_id GROUP BY u.user_id,u.full_name;
```

21. Top Feedback Providers

```
SELECT u.full_name, COUNT(f.feedback_id) AS feedbacks FROM Users u JOIN Feedback f ON  
u.user_id=f.user_id GROUP BY u.full_name ORDER BY feedbacks DESC LIMIT 5;
```

22. Duplicate Registrations Check

```
SELECT user_id,event_id,COUNT(*) AS duplicates FROM Registrations GROUP BY user_id,event_id  
HAVING COUNT(*) > 1;
```

23. Registration Trends

```
SELECT YEAR(registration_date) AS year, MONTH(registration_date) AS month, COUNT(*) AS  
registrations FROM Registrations WHERE registration_date >= CURDATE()-INTERVAL 12 MONTH  
GROUP BY YEAR(registration_date),MONTH(registration_date) ORDER BY year,month;
```

24. Average Session Duration per Event

```
SELECT e.title, AVG(TIMESTAMPDIFF(MINUTE,s.start_time,s.end_time)) AS avg_duration FROM  
Events e JOIN Sessions s ON e.event_id=s.event_id GROUP BY e.title;
```

25. Events Without Sessions

```
SELECT e.title FROM Events e LEFT JOIN Sessions s ON e.event_id=s.event_id WHERE s.session_id IS  
NULL;
```